Lesson 02 Demo 03

Creating a React App with Props and State Using Vite

Objective: To build a React app using Vite, demonstrating props and state through functional components, covering project setup, reusable components, passing data via props, and effective state management

Tools required: Visual Studio Code and Node.js

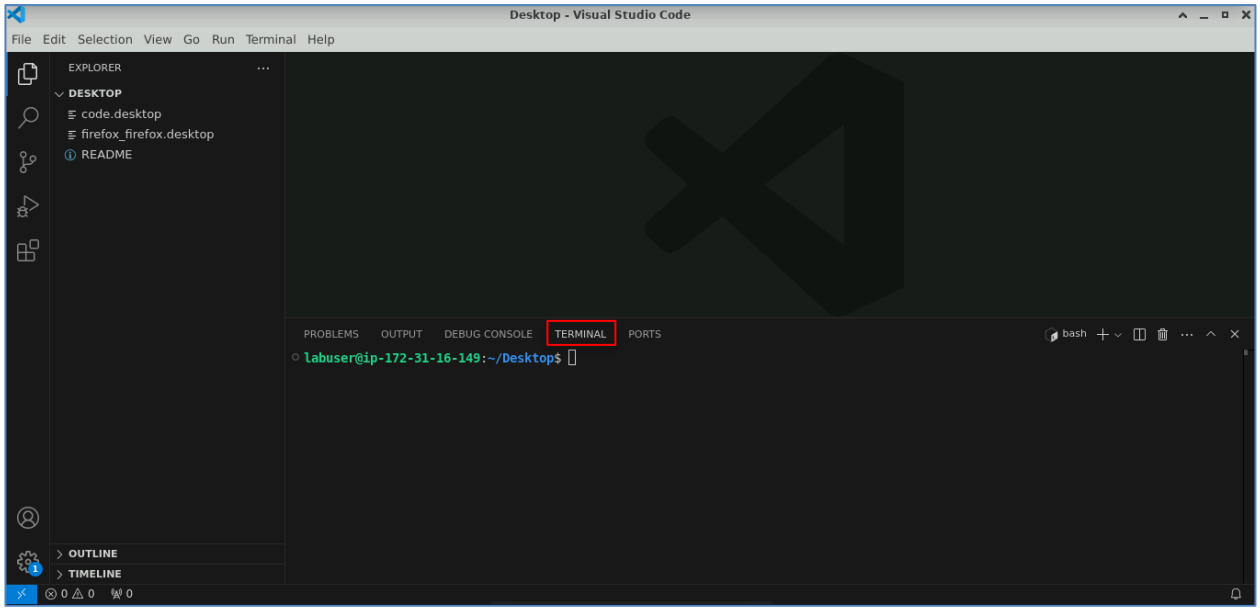
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create React components
3. Implement and use the component
4. Run and test the application

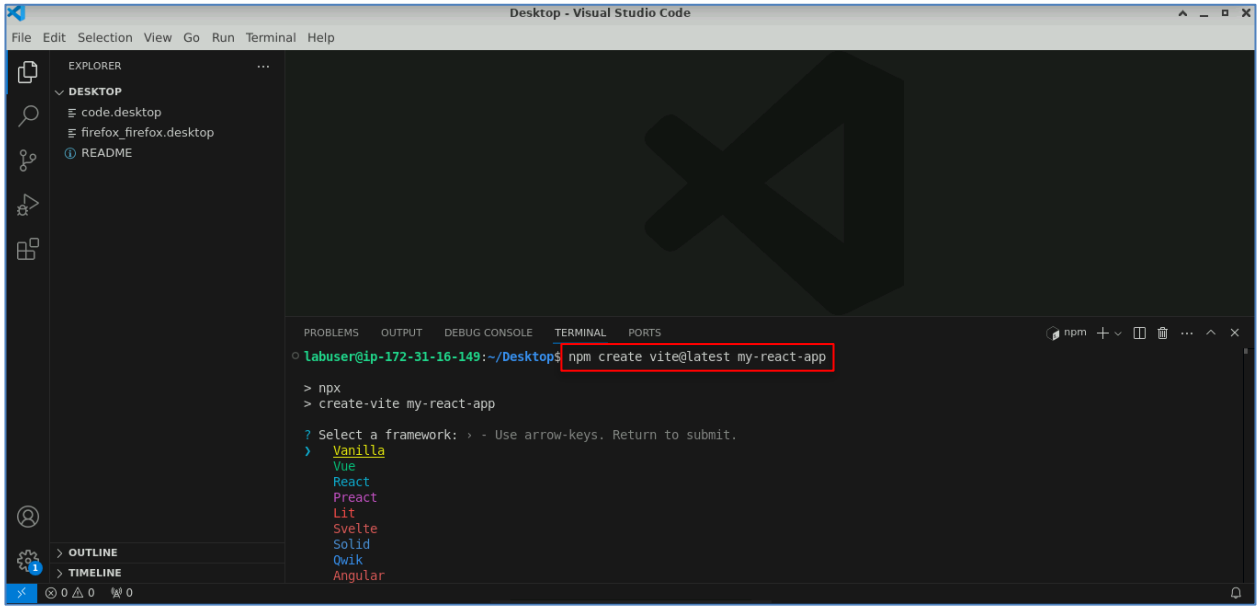
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

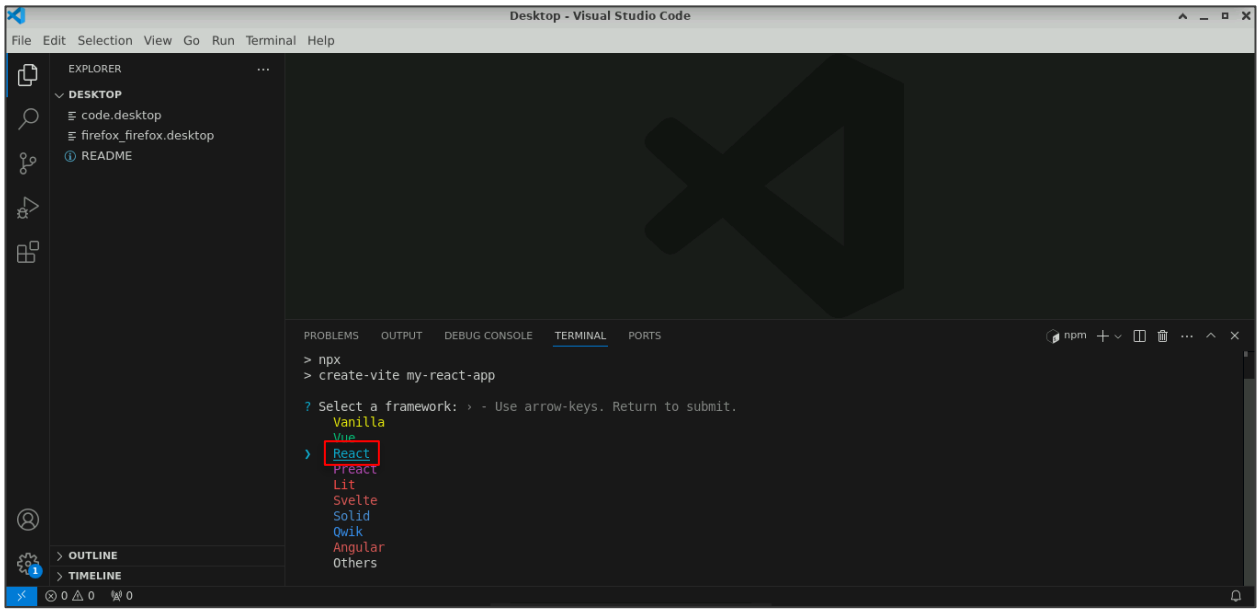


1.2 Run the following command to create a new React project using Vite:

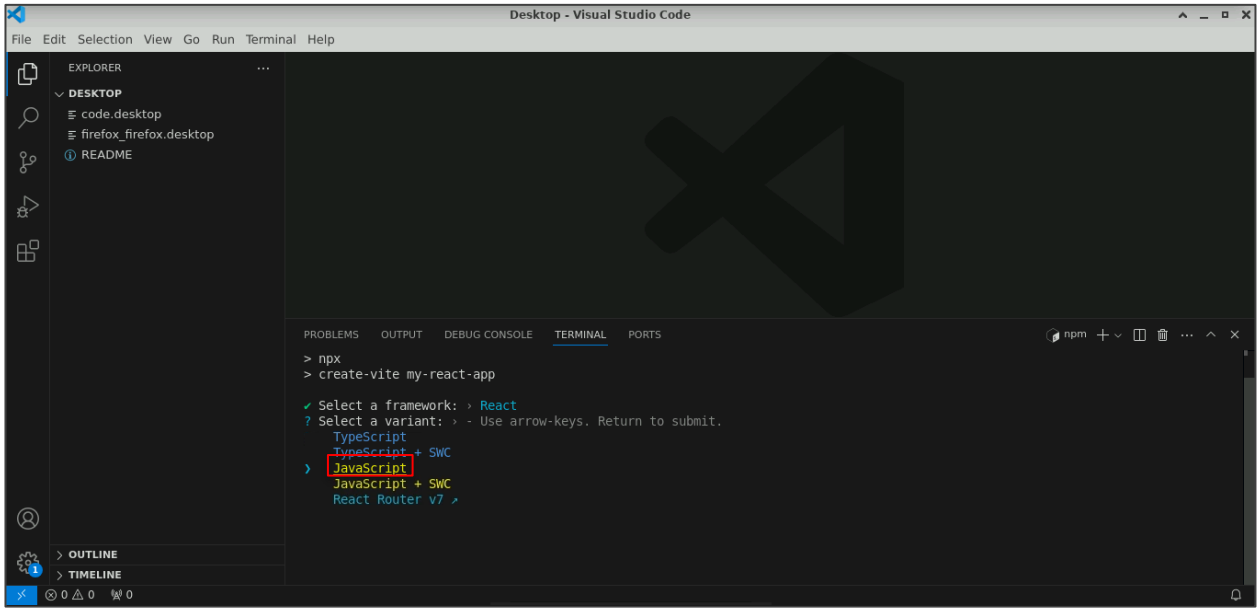
npm create vite@latest my-react-app



1.3 Select **React** and press enter

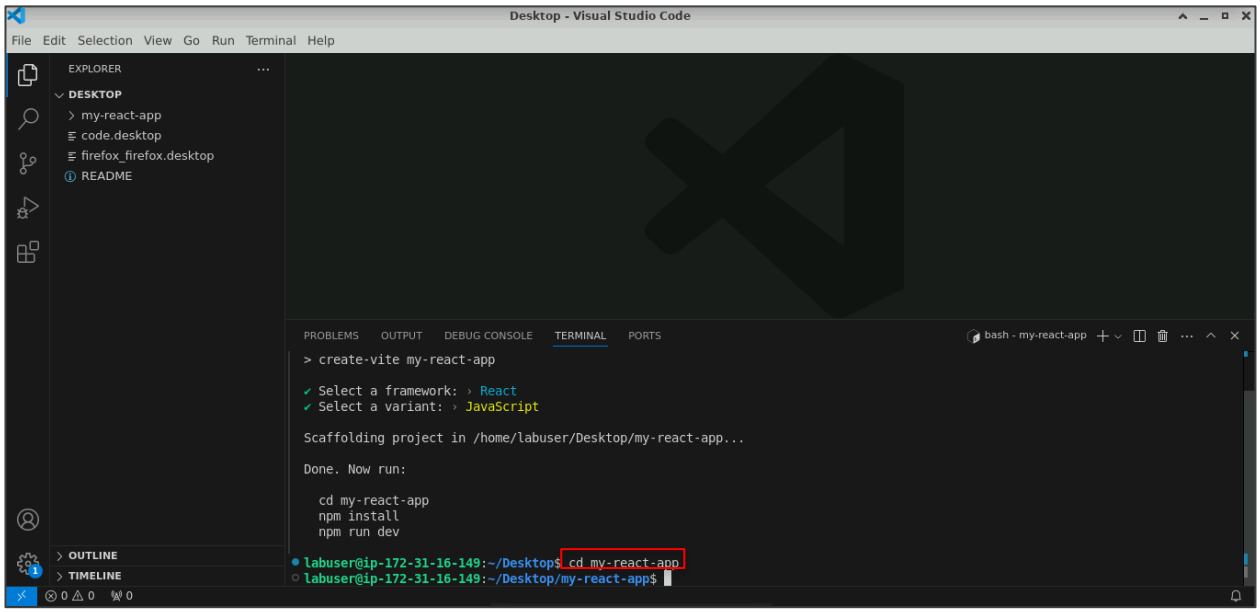


1.4 Select **JavaScript** and press enter



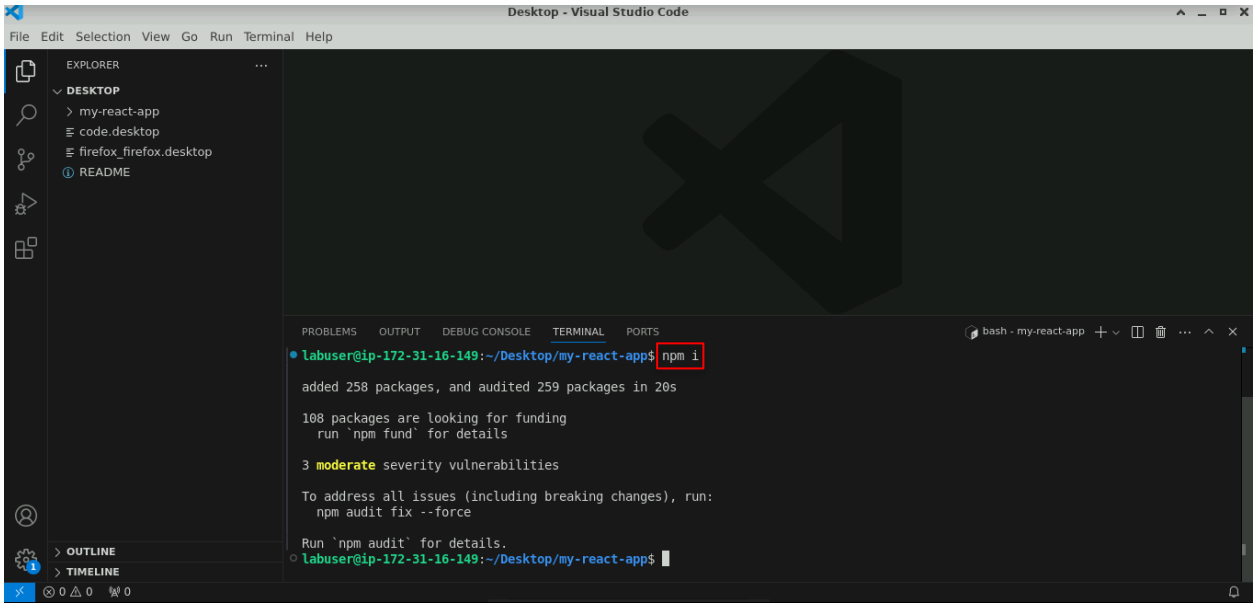
1.5 Run the following command to navigate to the project folder:

cd my-react-app



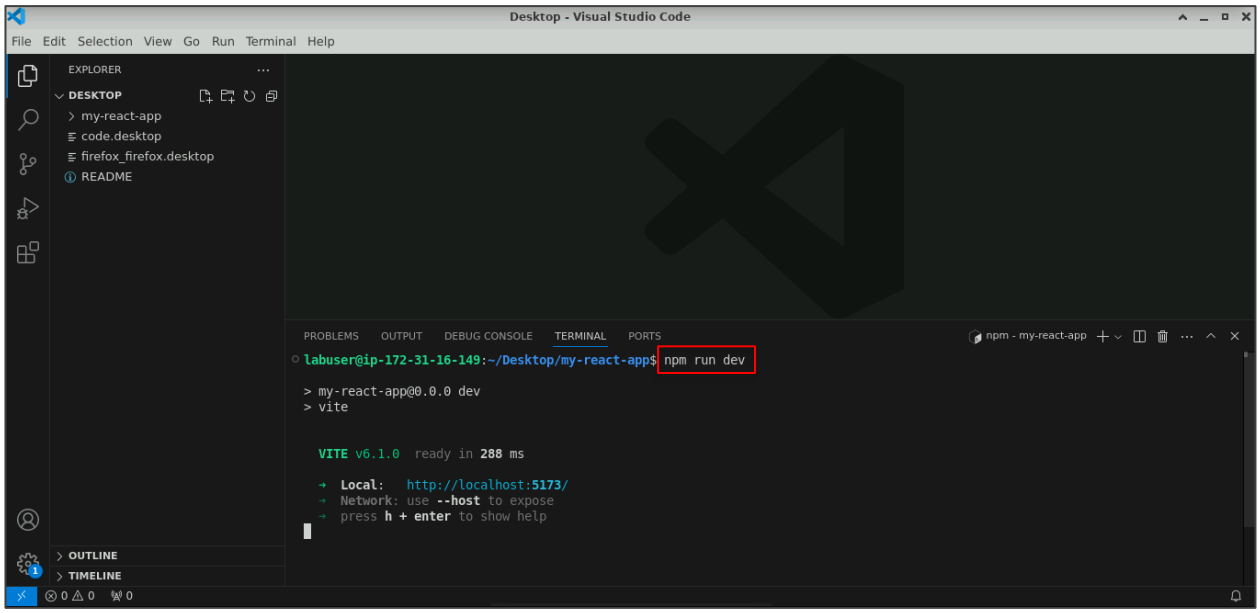
1.6 Run the following command to install dependencies:

npm i



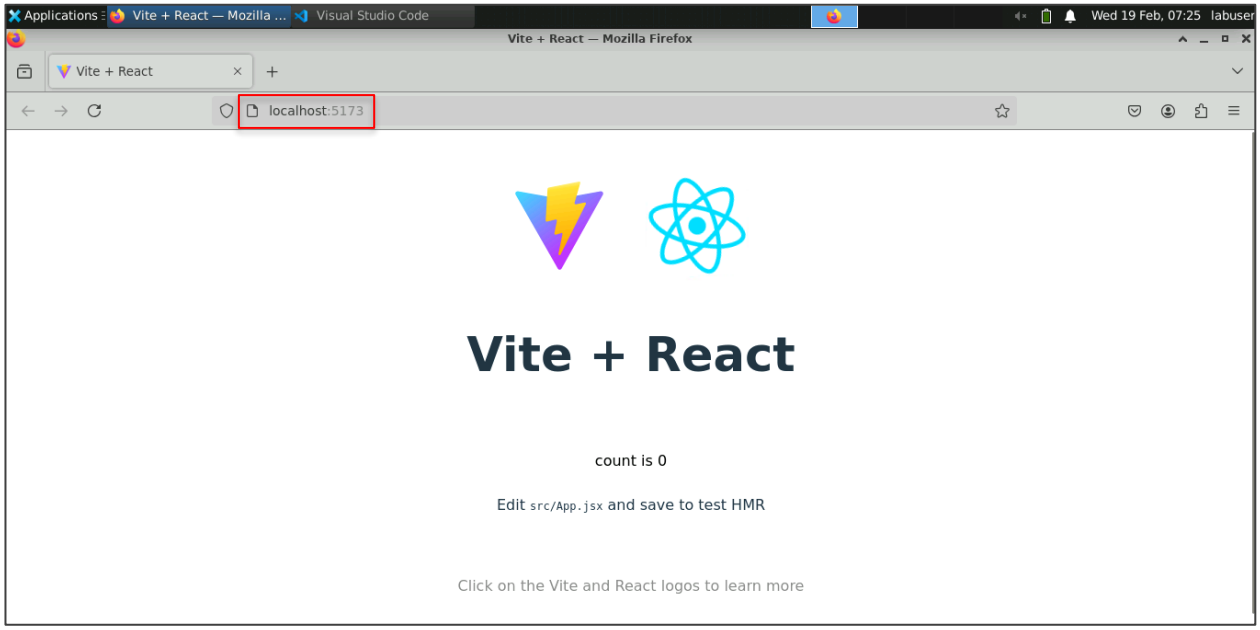
1.7 Run the following command to start the development server:

npm run dev



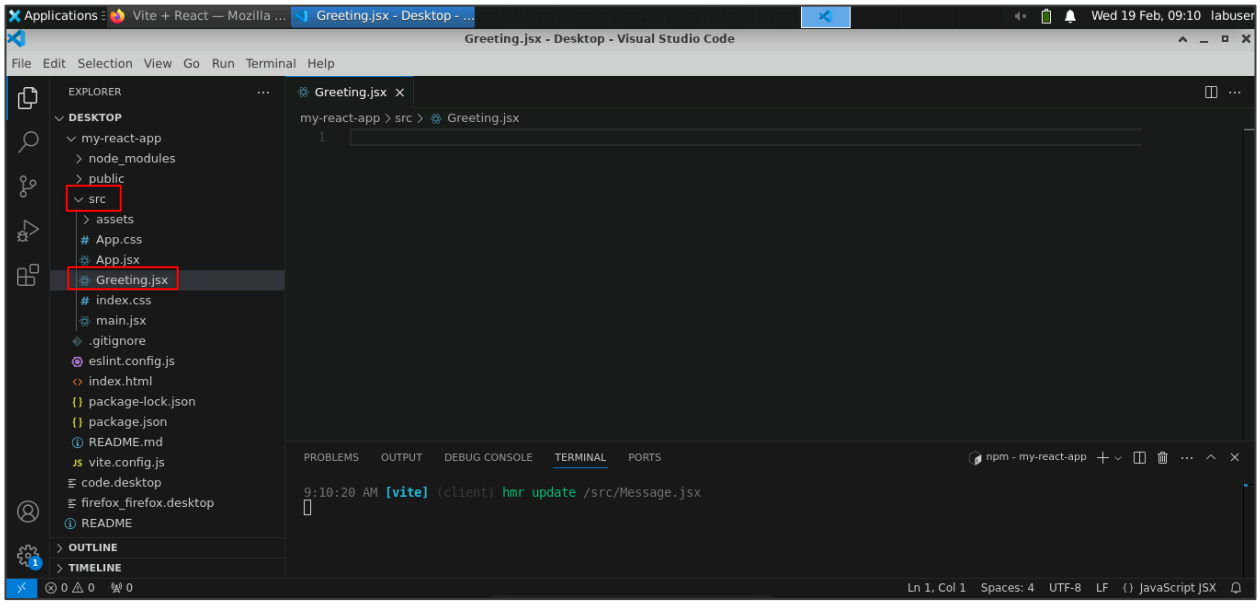
1.8 Navigate to the web browser and paste the link below:

http://localhost:5173



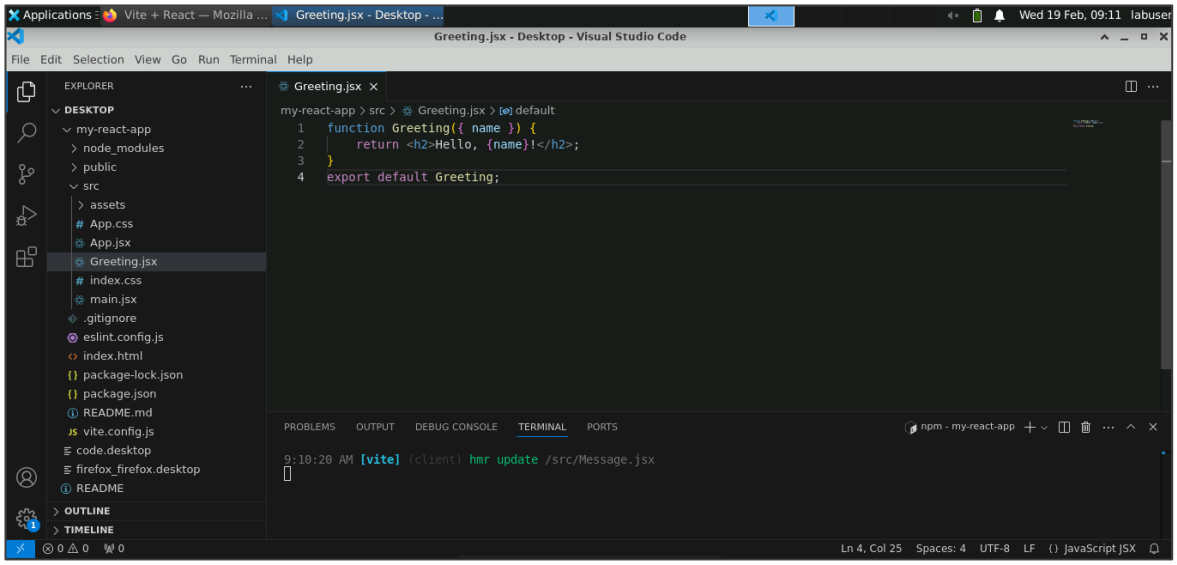
Step 2: Create React components

2.1 Inside the **src** folder, create a new file named **Greeting.jsx**

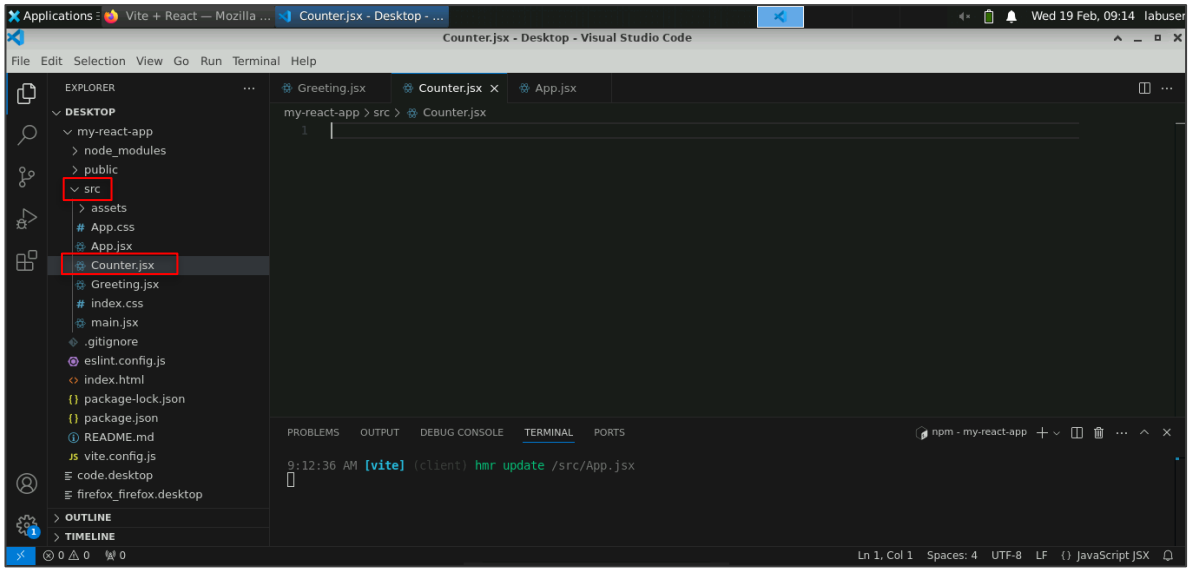


2.2 Paste the code below in **Greeting.jsx**:

```
function Greeting({ name }) {
   return <h2>Hello, {name}!</h2>;
}
export default Greeting;
```

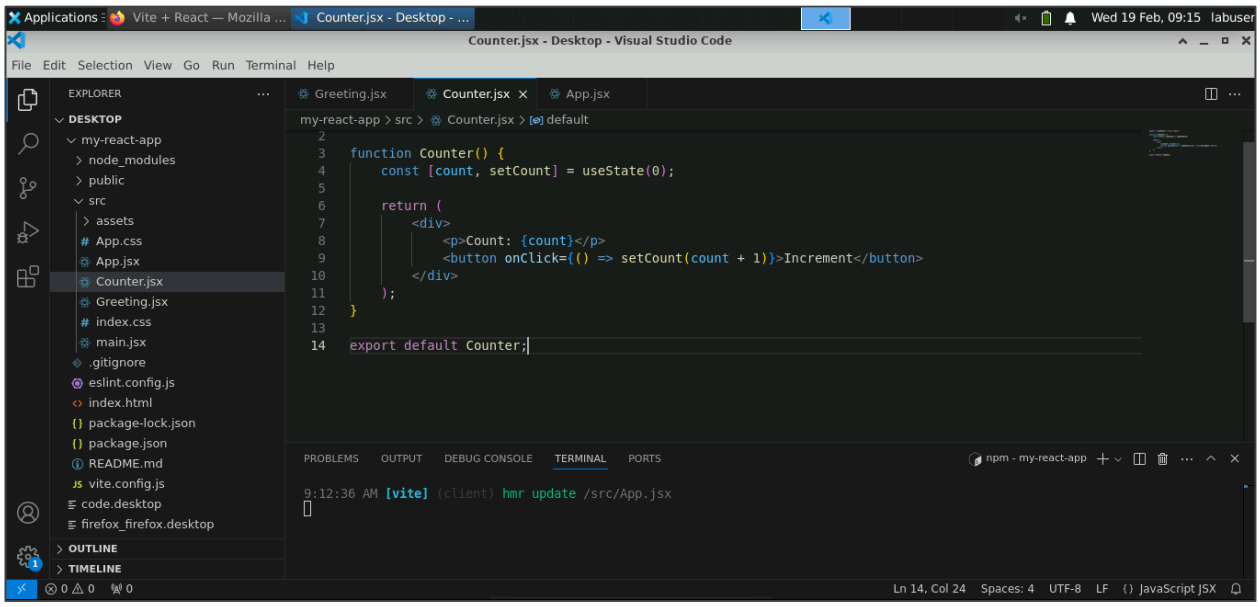


2.3 Inside the **src** folder, create a new file named **Counter.jsx**



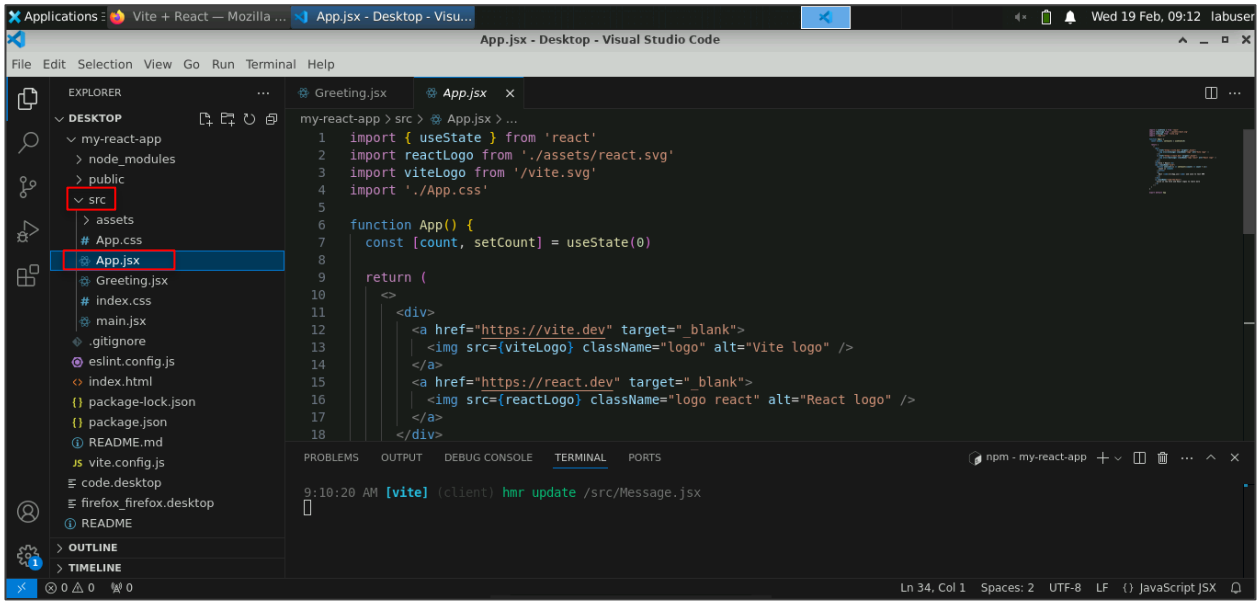
3.1 Paste the code below in **Counter.jsx**

```
import { useState } from "react";

function Counter() {
  const [count, setCount] = useState(0);

  return (
    <div>
      <p>Count: {count}</p>
      <button onClick={() => setCount(count + 1)}>Increment</button>
    </div>
  );
}

export default Counter;
```



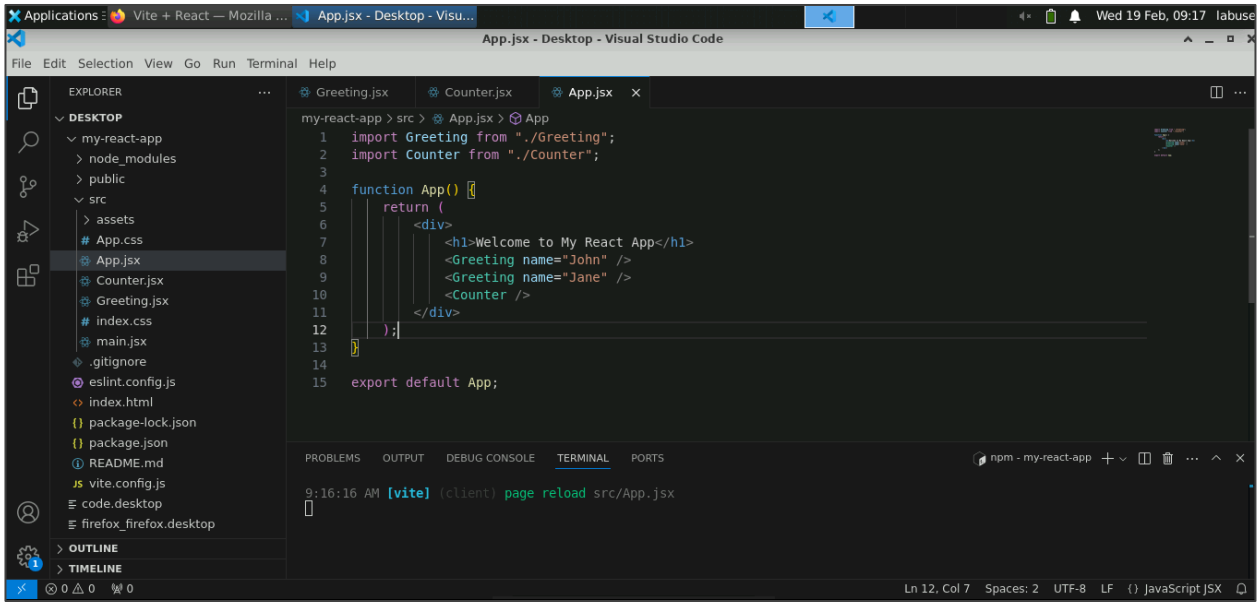
Step 3: Implement and use the component

3.2 Open the **App.jsx** file inside the **src** folder



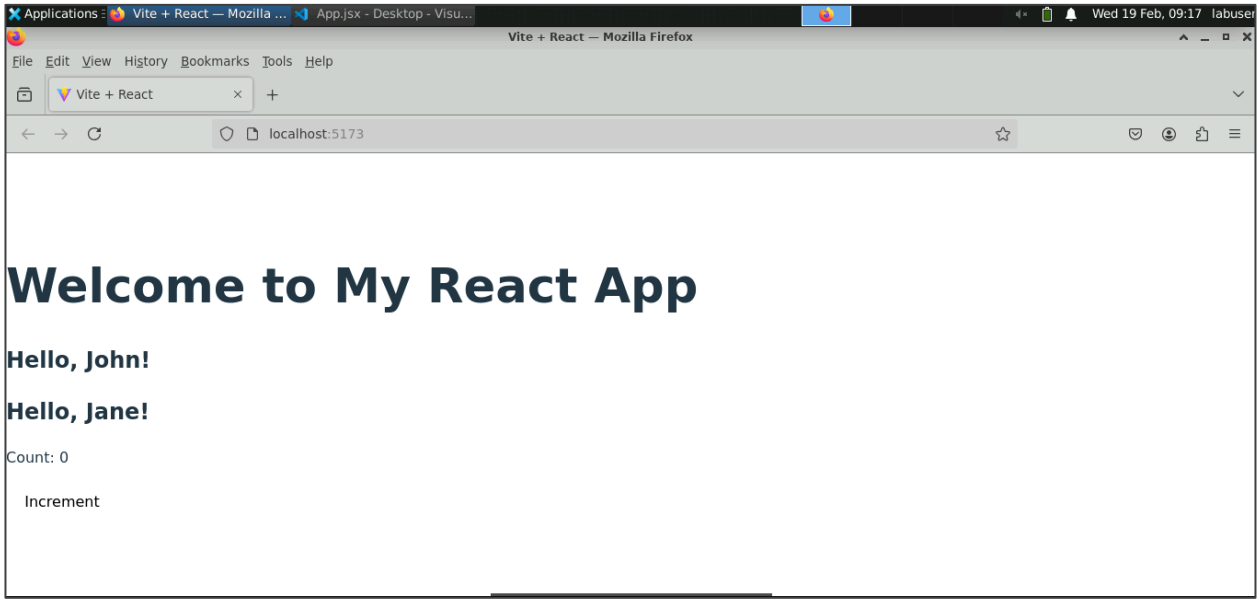
3.3 Paste the code below to include the new components in **App.jsx**:

```
import Greeting from "./Greeting";
import Counter from "./Counter";

function App() {
  return (
    <div>
      <h1>Welcome to My React App</h1>
      <Greeting name="John" />
      <Greeting name="Jane" />
      <Counter />
    </div>
  );
}

export default App;
```

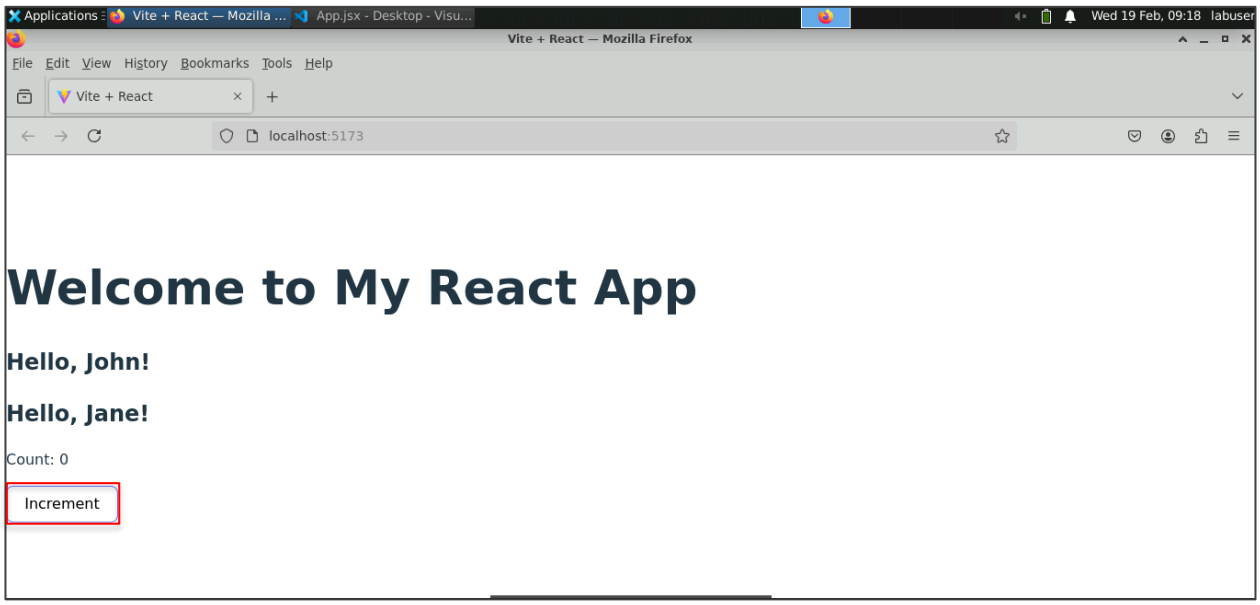


Step 4: Run and test the application

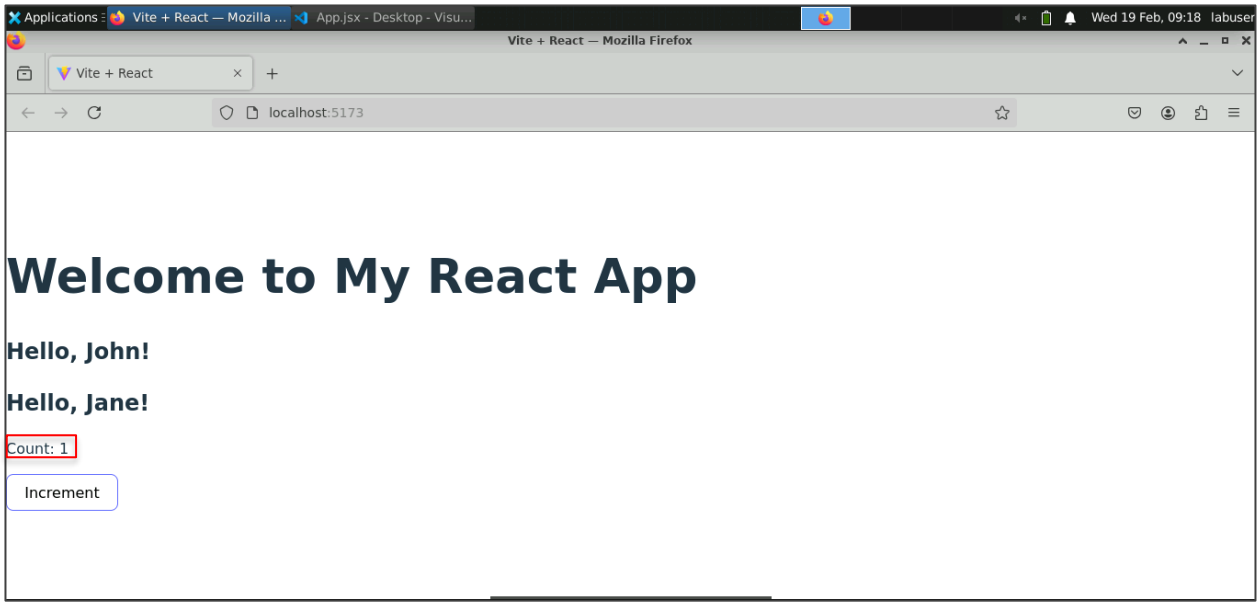
4.1 Navigate to the browser and refresh the webpage



4.2 Click **Increment** to update the Count



The **Count** will be incremented by one with each click.



By following the above steps, you have successfully created a React application using Vite, implemented reusable components, passed data via props, and managed the state effectively. The Greeting component demonstrates how to pass and display dynamic data using props, while the Counter component illustrates state management with the useState hook.